

ZAMZAM UNIVERSITY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

CARUURKAB AI: AN ARTIFICIAL INTELLIGENCE-BASED LEARNING PLATFORM FOR SOMALI CHILDREN

FINAL YEAR PROJECT REPORT

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DECLARATION

We hereby declare that this project report entitled "**Caruurkab AI: An Artificial Intelligence-Based Learning Platform for Somali Children**" is our original work. It has not been submitted to any other university or academic institution for the award of any degree or academic qualification before.

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ABSTRACT

Artificial Intelligence (AI) has emerged as a transformative technology in modern education, enabling personalized learning experiences that adapt to individual student needs. This report presents the design, development, and evaluation of **Caruurkab AI**, an intelligent educational mobile application specifically designed for Somali children aged 3 to 8 years.

The application integrates mobile learning technologies, speech-based interaction, and adaptive educational content to improve literacy, numeracy, and basic Islamic studies among early learners in their native language (Somali). The frontend of the system was built using Google's Flutter framework to ensure high performance and cross-platform compatibility. The backend utilizes Firebase cloud services for secure authentication, real-time database management, and file storage. AI algorithms were integrated to provide personalized learning paths, interactive quizzes, progress monitoring, and voice-assisted guidance.

Caruurkab AI directly addresses the severe lack of localized, high-quality educational apps for Somali-speaking children. It aims to transform mobile devices from simple entertainment platforms into productive educational assistants. A preliminary evaluation conducted within Mogadishu shows that the application significantly increases student engagement, focus, and knowledge retention, while providing parents with effective tools to monitor and control their child's digital activities.

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CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

In the contemporary digital era, Artificial Intelligence (AI) has become one of the fastest-growing technologies, deeply influencing healthcare, transportation, business, agriculture, and particularly education. Educational technology has undergone a major transformation through the integration of AI. Modern systems can now provide personalized learning paths, automated grading, intelligent tutoring, and adaptive educational content.

For children in their early stages of growth, learning plays a critical role in shaping cognitive abilities, communication skills, emotional intelligence, and future academic performance. Academic research shows that young children learn more effectively when educational content is visual, interactive, gamified, and delivered in their native language.

Despite global advancements in educational technology, Somalia still faces major challenges in accessing quality digital education. Many local schools suffer from a lack of teaching resources, qualified educators, and proper technological infrastructure. Furthermore, almost all digital educational platforms available online are designed in foreign languages, such as English or Arabic, which limits their accessibility to young Somali students.

Caruurkab AI is designed to bridge this educational gap. It serves as an AI-powered mobile learning platform specifically built for Somali-speaking children. The application aims to transform standard smartphones and tablets into intelligent learning assistants that deliver culturally relevant, language-friendly, and interactive learning experiences.

1.2 Statement of the Problem

Early childhood education in Somalia faces several technological and structural challenges, including:

- A severe lack of interactive, high-quality educational apps in the Somali language.
- Limited digital learning resources that reflect local cultural values and school curricula.
- Excessive unproductive screen time, where children use mobile devices primarily to watch entertainment videos.
- A complete lack of intelligent parental monitoring systems that allow parents to guide their child's digital activities.
- A lack of integration of Artificial Intelligence in Somali educational systems.

Because most available educational apps are in English or Arabic, Somali children cannot use them independently, often requiring constant parental guidance. Consequently, parents struggle to steer their children toward productive educational screen time. This project addresses the lack of a localized, intelligent, and culturally relevant educational platform that can convert standard mobile devices into effective learning tools.

1.3 Objectives of the Study

General Objective:

To design, develop, and evaluate an Artificial Intelligence-powered mobile learning platform specifically designed for Somali-speaking children.

Specific Objectives:

1. To develop a user-friendly, visually engaging mobile app interface optimized for early learners.
2. To integrate speech recognition and AI recommendation capabilities.
3. To provide personalized learning paths that adapt dynamically based on a child's performance.
4. To increase learning retention and engagement using gamification, animations, and sound effects.
5. To provide parents with analytical tools to track progress and control screen time.
6. To enrich localized digital educational content in the Somali language.

1.4 Research Questions

1. How can Artificial Intelligence be utilized to create personalized learning experiences for Somali-speaking children?
2. To what extent does Caruurkab AI improve learning outcomes, concentration, and engagement in young children?
3. What technical and computational challenges exist in developing localized AI educational applications for the Somali language?
4. How effective are parental monitoring systems in tracking child progress and regulating device usage?
5. How can mobile technology be transformed into a productive learning tool rather than a source of entertainment distraction?

1.5 Significance of the Study

For Students: Provides an interactive and culturally relevant learning platform that makes education fun and accessible in their mother tongue.

For Parents: Offers a safe, monitored digital space that guarantees screen time is spent productively on learning.

For Educators: Offers a model of how digital technology and AI tools can be integrated into early childhood classrooms in Somalia.

For Developers: Provides design guidelines and technical patterns for building localized educational software in low-resource languages.

1.6 Scope and Delimitations

The project focuses on children aged 3 to 8 years in Mogadishu, Somalia. The software is developed as a cross-platform mobile application built with Google's Flutter framework and backed by Firebase cloud services. The educational content focuses on basic literacy, numeracy, and introductory Islamic studies, taught entirely in Somali. Although the system supports offline caching, an active internet connection is required for cloud database synchronization and voice processing features.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature, theories, technologies, and previous research in Artificial Intelligence, educational platforms, mobile learning systems, and early childhood digital education. The literature review establishes the academic and technical foundation for the development of Caruurkab AI.

2.2 Concept of Artificial Intelligence (AI)

Artificial Intelligence (AI) refers to the capability of computer systems to perform tasks that typically require human intelligence, such as visual perception, decision-making, speech recognition, and translation. Driven by advancements in machine learning, deep neural networks, and cloud computing, AI has transitioned from research labs into daily practical use.

Unlike traditional software systems that follow static, predefined rules, AI systems can analyze user data, identify patterns, learn from experiences, and adapt their behavior dynamically. This capability is highly valuable in building adaptive learning environments.

2.3 Artificial Intelligence in Education (AIEd)

AI is transforming modern education by providing personalized educational environments. In traditional classrooms, a teacher delivers a uniform lesson to all students, regardless of their individual learning speeds. AIEd platforms solve this limitation by monitoring student performance and automatically tailoring lessons to match their strengths and weaknesses.

For instance, if a child struggles with a particular set of mathematics exercises, the system provides supplementary visual hints and simpler problems. Once the child masters the concept, the system increases the difficulty. AI technologies also power automated assessments, voice assistants, and learning analytics that keep parents and teachers updated.

2.4 Mobile Learning Systems

Mobile learning (m-learning) is the use of portable devices, such as smartphones and tablets, to access learning content. M-learning is highly flexible, allowing kids to learn at their own pace, outside formal school hours. It is particularly effective for young children because of their natural interest in touchscreen interactions, colorful visuals, and audio-visual animations.

2.5 Early Childhood Education and Technology

During the critical developmental years of 3 to 8, children form foundational cognitive, motor, and linguistic skills. Research in child development indicates that educational technology is most beneficial when it uses simple, child-friendly user interfaces, colorful visuals, positive reinforcement sounds, and gamified tasks. The addition of AI allows the app to dynamically guide the child without making the interface overly complex.

2.6 Educational Technology in Somalia

Somalia's education sector continues to face challenges, including a lack of physical textbooks, school facilities, and trained teachers. While mobile internet and smartphone ownership have increased rapidly, local educational software in the Somali language remains nearly non-existent. Most available applications are in English or Arabic, which presents a significant barrier for young children. Caruurkab AI addresses this gap by offering a culturally relevant platform in the Somali language.

2.7 Related Educational Platforms

Duolingo: Excellent gamification and adaptive learning model, but lacks support for Somali language instruction.

Khan Academy: Excellent video library and tests, but is not localized for early Somali education and requires fast internet.

Google Classroom: Strong management tools, but designed for formal school environments and older users, not early childhood.

YouTube Kids: Offers vast video content, but focuses on passive video consumption and lacks interactive educational evaluation.

2.8 Theoretical Framework

This project is guided by two main educational theories:

1. **Constructivist Learning Theory:** Proposes that children learn by exploring, testing, and interacting. Caruurkab AI supports this through interactive games, drag-and-drop puzzles, and audio feedback.
2. **Personalized Learning Theory:** Emphasizes that instruction should be tailored to each student's speed and learning style. AI algorithms in this app monitor quiz results to suggest appropriate learning paths.

2.9 Research Gap

While global educational apps are highly advanced, there is a clear gap in digital tools built in the Somali language. Additionally, existing systems do not reflect Somali cultural values, nor do they offer parental monitoring tools adapted to Somali households. Caruurkab AI is designed to fill this educational and technological gap.

2.10 Chapter Summary

This chapter reviewed research on AI in education, mobile learning, child psychology, and the local educational challenges in Somalia. The findings confirm that a localized, AI-driven educational mobile app is highly needed to support Somali children's early education.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Introduction

This chapter outlines the research design, data collection methods, software development process, and tools used to develop the application.

3.2 Research Design

The project combined qualitative and quantitative methods. Qualitative methods were used to understand user needs, while quantitative methods evaluated student performance and user engagement levels.

3.3 Data Collection Methods

- **Interviews:** Conducted with parents and early childhood teachers in Mogadishu to gather requirements.
- **Observations:** Observed children interacting with digital devices to design a simple, child-friendly user interface.
- **Surveys:** Distributed questionnaires to evaluate parental satisfaction and ease of use.

3.4 Software Development Methodology

The software was built using the **Agile Methodology**. This approach allowed the team to work in short cycles (sprints) to plan, design, implement, test, and refine features. This iterative model allowed us to gather feedback from child testing sessions and implement changes quickly.

3.5 Tools and Technologies

- **Flutter & Dart:** Used to build the cross-platform mobile frontend interface.
- **Firebase:** Used for backend cloud database storage, media asset hosting, and secure user authentication.
- **Supabase:** Integrated for structured SQL data storage, progress management, and notifications.
- **VS Code & Android Studio:** Primary development environment IDEs.

CHAPTER FOUR: SYSTEM ANALYSIS, DESIGN AND IMPLEMENTATION

4.1 Introduction

This chapter describes the functional requirements, system architecture, database design, user interface principles, implementation details, testing procedures, and deployment strategies.

4.2 Functional Requirements

- **User Authentication:** Secure registration and login for parents and students.
- **Interactive Lessons:** Delivery of multimedia educational content (Somali, English, Math, Saynis).
- **Dynamic Quizzes:** Automatic generation of MCQ, True/False, and Fill-in-the-Blank questions.
- **Progress Tracking:** Real-time scores and lesson completion saved in the cloud.
- **AI Recommendation:** Auto-recommending lessons based on learning speed.
- **Parent Dashboard:** Progress charts and device control tools.

4.3 Non-Functional Requirements

- **Security:** User credentials and progress data must be encrypted in transit and at rest.
- **Performance:** Touch response time must be under 150 milliseconds to keep children engaged.
- **Usability:** The interface must be visually intuitive, using minimal text and maximum audio-visual cues.

4.4 System Architecture

Caruurkab AI utilizes a 3-tier client-server architecture consisting of:

1. **Frontend Client Layer:** The Flutter mobile app, running on Android devices.
2. **Cloud Integration & Database Layer:** Powered by Firebase (auth, media storage) and Supabase (relational SQL data).
3. **AI Processing Layer:** Handles speech recognition, chatbot queries, and personalized learning algorithms.

4.5 Use Case Diagram & Database Design

The system identifies three primary users: **Student** (completes lessons, takes quizzes, gets AI suggestions), **Parent** (registers the child, views progress reports, receives notifications), and **Administrator** (manages educational courses, edits quizzes).

The database schema consists of several interconnected tables: **users** (profile details, school class levels), **lessons** (subject chapters, text content), **quizzes** (JSON-formatted question sheets, correct answer indices), **lesson_progress** (timestamps of completed tasks), and **student_quiz_progress** (high scores, earned badges).

4.6 System Implementation

The mobile app's screens were developed using Flutter's widget tree. We implemented advanced learning progression features. For example, lesson screens dynamically query database progress to ensure students complete tasks in a logical, step-by-step sequence. Furthermore, we added a complete **Fill-in-the-Blank** quiz parser that shuffles three options and displays them below the question, allowing the student to visually select an option that automatically inserts into the blank space.

On the backend, security rules were configured to block unauthorized users. We also implemented database functions to update the status of lessons in a chapter once the student passes the chapter quiz.

Sample Dart Source Code: Firebase Email Login

```
await FirebaseAuth.instance.signInWithEmailAndPassword(  
  email: emailController.text.trim(),  
  password: passwordController.text.trim(),  
);
```

4.7 System Testing and Evaluation

Testing was carried out throughout the development process. Unit tests validated database reads and login functions, integration tests ensured cloud storage and database APIs worked together smoothly, and user acceptance testing evaluated child engagement and parental satisfaction.

Test Case	Expected Output	Status
User Registration	Profile created in database	Pass
Quiz Evaluation	Score generated; progress stored	Pass
Fill-in-the-Blank Options	3 choices rendered; blank updated	Pass
Parent Dashboard Charts	Accurate performance charts shown	Pass

4.8 Deployment and Maintenance

The application was compiled into a signed release APK and AppBundle (AAB) to prepare for Google Play Store deployment. Database access controls were updated, and backend server functions were optimized. Future maintenance will include regular security updates, database cleanups, and performance improvements.

CHAPTER FIVE: CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

Caruurkab AI successfully demonstrates that Artificial Intelligence and mobile learning technologies can make quality education highly accessible and engaging for Somali children. By designing a culturally friendly interface in their native language, the application transforms standard mobile devices into effective, interactive learning assistants.

The system's features (such as personalized lesson recommendations, interactive chapter quizzes, and progress tracking) provide a solid base for digital learning in Somalia. The project proves that localized software solutions can significantly improve literacy, numeracy, and child focus.

5.2 Recommendations

1. **Offline Learning Mode:** Support local databases for lesson caching to help students with weak internet.
2. **Video Learning:** Add short, interactive cartoon videos to make complex subjects easier.
3. **School Integration:** Partner with local primary schools to deploy the app as a classroom learning tool.
4. **Advanced Speech AI:** Improve speech recognition to help young kids practice pronunciation.
5. **Expanding Curriculum:** Add additional courses aligned with the national Somali primary curriculum.

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APPENDICES

Appendix A: User Manual

For Parents:

- 1. Open the application and register a parent account.
- 2. Add your child's name and select their school class level (Fasalka 1 - 4).
- 3. Open the Parent Dashboard to monitor daily progress, check completed quizzes, and view progress charts.

For Students:

- 1. Select a subject (e.g., Af Soomaali) from the student dashboard.
- 2. Follow the learning chapters in order. Complete each lesson's exercises to unlock the next lesson.
- 3. Take the chapter quiz at the end of each module to unlock the next chapter.

Appendix B: Project Budget

Expense Item	Estimated Cost (USD)
Internet Services	\$50
Cloud Hosting (Firebase & Supabase)	\$40
Testing Devices (Android Tablets)	\$200
Software Subscriptions & Tools	\$30
Local Transportation	\$60
Miscellaneous	\$20
Total Budget	\$400

Appendix C: Project Timeline

Project Phase	Duration
Planning & Proposal Draft	2 Weeks
System Analysis & Requirements	2 Weeks
UI/UX & Database Design	3 Weeks
Mobile & Backend Development	5 Weeks
Testing & Bug Fixing	2 Weeks
Final Deployment & Documentation	1 Week